

Exercícios para a P1

[Elon]

3.2. Mostre que as matrizes **a**, **b** e **c** abaixo são L.I.:

$$\mathbf{a} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Pergunta adicional: Qual a dimensão do espaço gerado por **a**, **b** e **c**?

[Schaum]

4.101. Determine whether each of the following is a basis of the vector space $\mathbf{P}_n(t)$:

- (a) $\{1, \quad 1+t, \quad 1+t+t^2, \quad 1+t+t^2+t^3, \quad \dots, \quad 1+t+t^2+\dots+t^{n-1}+t^n\};$
- (b) $\{1+t, \quad t+t^2, \quad t^2+t^3, \quad \dots, \quad t^{n-2}+t^{n-1}, \quad t^{n-1}+t^n\}.$

Obs.: $\mathbf{P}_n(t)$ é o espaço dos polinômios em t de grau menor ou igual a n .

[Schaum]

EXAMPLE 5.2 Let V be the vector space of polynomials over $\mathbf{R}$, and let $p(t) = 3t^2 - 5t + 2$.

(a) The derivative defines a mapping $\mathbf{D}: V \rightarrow V$ where, for any polynomials $f(t)$, we have $\mathbf{D}(f) = df/dt$. Thus,

$$\mathbf{D}(p) = \mathbf{D}(3t^2 - 5t + 2) = 6t - 5$$

(b) The integral, say from 0 to 1, defines a mapping $\mathbf{J}: V \rightarrow \mathbf{R}$. That is, for any polynomial $f(t)$,

$$\mathbf{J}(f) = \int_0^1 f(t) dt, \quad \text{and so} \quad \mathbf{J}(p) = \int_0^1 (3t^2 - 5t + 2) = \frac{1}{2}$$

Observe that the mapping in (b) is from the vector space V into the scalar field $\mathbf{R}$, whereas the mapping in (a) is from the vector space V into itself.

[Schaum]

EXAMPLE 6.1 Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be the linear operator defined by $F(x, y) = (2x + 3y, 4x - 5y)$.

(a) Find the matrix representation of F relative to the basis $S = \{u_1, u_2\} = \{(1, 2), (2, 5)\}$.

- (1) First find $F(u_1)$, and then write it as a linear combination of the basis vectors u_1 and u_2 . (For notational convenience, we use column vectors.) We have

$$F(u_1) = F\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ -6 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and} \quad \begin{aligned} x + 2y &= 8 \\ 2x + 5y &= -6 \end{aligned}$$

Solve the system to obtain $x = 52$, $y = -22$. Hence, $F(u_1) = 52u_1 - 22u_2$.

- (2) Next find $F(u_2)$, and then write it as a linear combination of u_1 and u_2 :

$$F(u_2) = F\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 19 \\ -17 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and} \quad \begin{aligned} x + 2y &= 19 \\ 2x + 5y &= -17 \end{aligned}$$

Solve the system to get $x = 129$, $y = -55$. Thus, $F(u_2) = 129u_1 - 55u_2$.

Now write the coordinates of $F(u_1)$ and $F(u_2)$ as columns to obtain the matrix

$$[F]_S = \begin{bmatrix} 52 & 129 \\ -22 & -55 \end{bmatrix}$$

[Schaum]

- (b) Find the matrix representation of F relative to the (usual) basis $E = \{e_1, e_2\} = \{(1, 0), (0, 1)\}$.

Find $F(e_1)$ and write it as a linear combination of the usual basis vectors e_1 and e_2 , and then find $F(e_2)$ and write it as a linear combination of e_1 and e_2 . We have

$$\begin{aligned} F(e_1) = F(1, 0) &= (2, 2) = 2e_1 + 4e_2 \\ F(e_2) = F(0, 1) &= (3, -5) = 3e_1 - 5e_2 \end{aligned} \quad \text{and so} \quad [F]_E = \begin{bmatrix} 2 & 3 \\ 4 & -5 \end{bmatrix}$$

Note that the coordinates of $F(e_1)$ and $F(e_2)$ form the columns, not the rows, of $[F]_E$. Also, note that the arithmetic is much simpler using the usual basis of $\mathbf{R}^2$.

- (c) Utilize o Python para notar que $[F]_S$ e $[F]_E$ tem os mesmos autovalores

[Schaum]

7.11. Let W be the subspace of $\mathbf{R}^5$ spanned by $u = (1, 2, 3, -1, 2)$ and $v = (2, 4, 7, 2, -1)$. Find a basis of the orthogonal complement $W^\perp$ of W .

We seek all vectors $w = (x, y, z, s, t)$ such that

$$\langle w, u \rangle = x + 2y + 3z - s + 2t = 0$$

$$\langle w, v \rangle = 2x + 4y + 7z + 2s - t = 0$$

Eliminating x from the second equation, we find the equivalent system

$$x + 2y + 3z - s + 2t = 0$$

$$z + 4s - 5t = 0$$

The free variables are y, s , and t . Therefore,

(1) Set $y = -1, s = 0, t = 0$ to obtain the solution $w_1 = (2, -1, 0, 0, 0)$.

(2) Set $y = 0, s = 1, t = 0$ to find the solution $w_2 = (13, 0, -4, 1, 0)$.

(3) Set $y = 0, s = 0, t = 1$ to obtain the solution $w_3 = (-17, 0, 5, 0, 1)$.

The set $\{w_1, w_2, w_3\}$ is a basis of $W^\perp$.